Class Activity # 1
Calculus Physics 2

'ame: Key Date: _____

Group Members: _____ _____ _____

Solve the following problems together as a group, stay engaged in the conversation with the group!

- Use **Lecture Notes Chapter 34 and 35** as a guide to answer the following questions.

1) Which statement(s) are true about electric charge? (There can be more than one correct statement)

A B F G

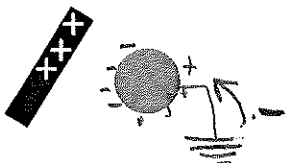
(A) The source of the positive charge is the proton.
(B) The source of the negative charge is the electron.
(C) The source of the positive charge can be either the proton or the electron.
(D) The source of the negative charge can be either the proton or the electron.
(E) The magnitude of the charge is the same for both the proton and the electron.
(F) It is possible to have a net charge of 1×10^{-19} C
(G) It is possible to have a net charge of 3.2×10^{-19} C
(H) An electron has a charge of -1.6×10^{-19} C and the proton has a charge of $+1.6 \times 10^{-19}$ C.

2) What is the type and the amount of net charge for a carbon atom when three electrons are removed?

$Q = N q_e = 3(1.6 \times 10^{-19}\ C) = 4.8 \times 10^{-19}$ positive

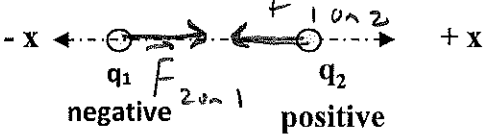
3) A plastic rod with a net positive charge is brought close to (but does not touch) a neutral metal conducting sphere that is connected to ground. After waiting a few seconds, the ground connection is removed (without touching the sphere), and after that the rod is also removed. The sphere is now

(A) negatively charged.
B) positively charged.
C) neutral.



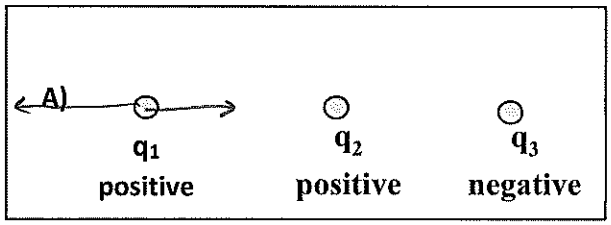
4) Two charge objects are separated by a distance r, $q_1 = 10\ \mu C$ and is negative, $q_2 = 20\ \mu C$ and is positive. Which statement(s) is true concerning the two charged objects shown? (There can be more than one true statement)?

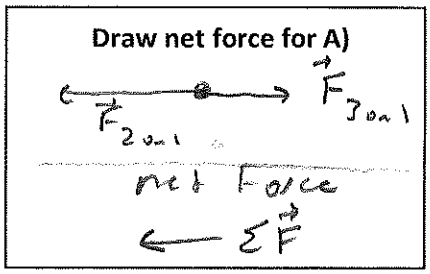
C D G

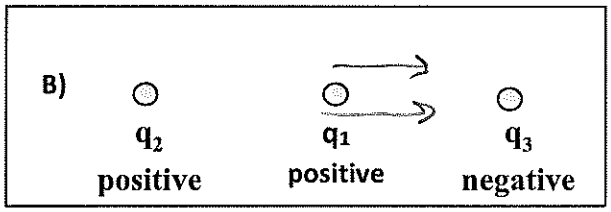


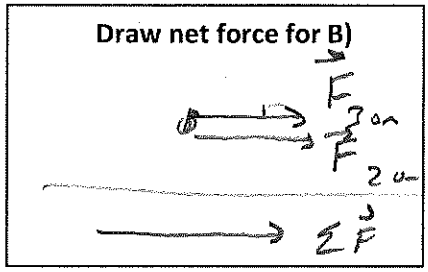
A) The force acting on the negative charged object q_1 is in the negative direction.
B) The force acting on the positive charged object q_2 is in the positive direction.
(C) The force acting on the negative charged object q_1 is in the positive direction.
(D) The force acting on the positive charged object q_2 is in the negative direction.
E) The force acting on object 1 is twice as much as the force acting on object 2.
F) The force acting on object 2 is twice as much as the force acting on object 1.
(G) The force of object 1 is equal and opposite to the force of object 2 on object 1.

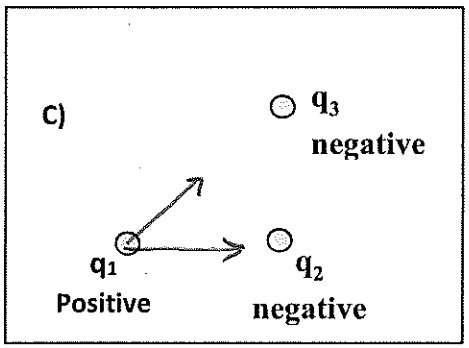
5) Draw all the electric forces acting on the **sphere #1** and the net force vector for each of the situations below and label, assume the magnitude of the net charge is the same for all spheres.

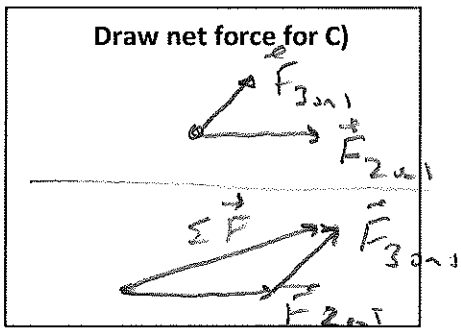






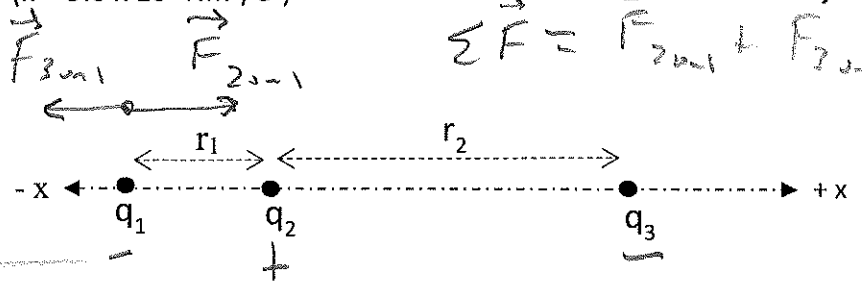






6) Three point charges are placed on the *x*-axis. Determine the magnitude and direction of the electrostatic force which acts on charge q_1? ($k = 9.0 \times 10^9\ Nm^2/C^2$)

$q_1 = 10\ \mu C$ negative
$q_2 = 20\ \mu C$ positive
$q_3 = 30\ \mu C$ negative
$r_1 = 0.5$ m
$r_2 = 1.0$ m



$\vec{\Sigma F} = \vec{F}_{2on1} + \vec{F}_{3on1}$

$|\vec{F}_{2on1}| = \frac{k q_2 q_1}{r_1^2} = 7.2\ N$

$|\vec{F}_{3on1}| = \frac{k q_3 q_1}{(r_1 + r_2)^2} = 1.2\ N$

$\vec{F}_{2on1} = (+7.2N)\hat{x} + (0)\hat{y}$

$\vec{F}_{3on1} = (-1.2N)\hat{x} + (0)\hat{y}$

$\vec{\Sigma F} = (+6N)\hat{x} + (0N)\hat{y}$

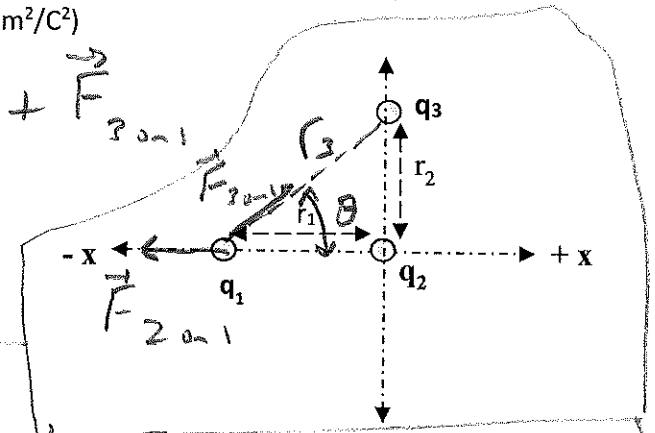
$|\vec{\Sigma F}| = 6N$

$\alpha = 0°$ $\beta = 90°$

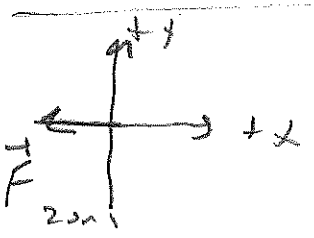
7) Determine the magnitude and direction of the net electrostatic force acting on point charge q_1 given the magnitude and type of charge given below. ($k = 9.0 \times 10^9\ Nm^2/C^2$)

$q_1 = 50\ \mu C$ negative
$q_2 = 50\ \mu C$ negative
$q_3 = 50\ \mu C$ positive
$r_1 = 0.8\ m$
$r_2 = 0.8\ m$

$$\Sigma \vec{F} = \vec{F}_{2on1} + \vec{F}_{3on1}$$



Need Vectors in Component Form

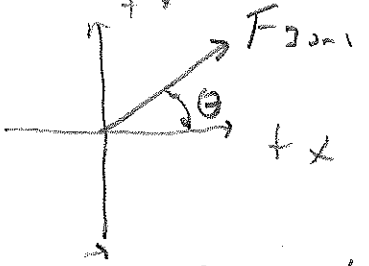


$$\vec{F}_{2on1} \cdot \hat{x} = |\vec{F}_{2on1}| \cos(\alpha)$$

$$\alpha = 180°$$

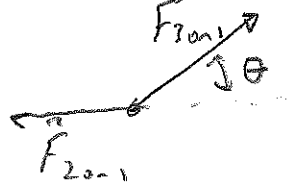
$$F_{2on1}(x) = -35.2\ N$$

$$\vec{F}_{2on1} = (-35.2\ N)\hat{x} + (0\ N)\hat{y}$$



$$\alpha = 45° = \theta$$

$$\beta = 90° - \theta = 45°$$



$$r_3 = \sqrt{r_1^2 + r_2^2} = 1.13\ m$$

$$\theta = \tan^{-1}\left(\frac{r_2}{r_1}\right) = 45°$$

$$F_{3on1} \cdot \hat{x} = |\vec{F}_{3on1}| \cos(\alpha) = 12.5\ N$$

$$\vec{F}_{3on1} \cdot \hat{y} = |\vec{F}_{3on1}| \cos(\beta) = 12.5\ N$$

$$\vec{F}_{3on1} = (12.5\ N)\hat{x} + (12.5\ N)\hat{y}$$

$$|\vec{F}_{2on1}| = \frac{k q_2 q_1}{r_1^2} = 35.2\ N$$

$$|\vec{F}_{3on1}| = \frac{k q_3 q_1}{r_3^2} = 17.6\ N$$

$$\Sigma \vec{F} = (\Sigma F_x)\hat{x} + (\Sigma F_y)\hat{y}$$

$$\Sigma F_x = -35.2\ N + 12.5\ N = -22.7\ N$$

$$\Sigma F_y = 0\ N + 12.5\ N = +12.5\ N$$

$$\Sigma \vec{F} = (-22.7\ N)\hat{x} + (12.5\ N)\hat{y}$$

$$|\Sigma F| = \sqrt{\Sigma F_x^2 + \Sigma F_y^2} = 25.9\ N$$

$$\alpha = \cos^{-1}\left(\frac{\Sigma F_x}{|\Sigma F|}\right) = 151°$$

$$\beta = \cos^{-1}\left(\frac{\Sigma F_x}{|\Sigma F|}\right) = 61°$$